

Predicting Emergency Room Wait Times Using Machine Learning

From weak baseline results to a stronger, leakage-aware predictive model



The ER Challenge: Can We Predict the Wait?

Long emergency room waits strain resources, frustrate patients, and impact care quality. Machine learning offers a potential solution to a critical healthcare problem.



Project Objective & Research Question



Predict ER wait times using machine learning.



Evaluate useful features and methods.



Which factors best explain ER wait times?



How reliably can ML predict them?



Hypothesis: Operational factors outperform demographics.

Our ML Workflow: An Iterative Journey

From raw data to refined insights



Data Cleaning & Feature Engineering

Prepare and enrich data for modeling.



Train/Test Splitting & Baseline Modeling

Establish a performance benchmark.



Regression Models & Error Analysis

Experiment with various predictive models.



Feature Analysis & Refinement

Understand feature impact and model behavior.





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Initial Attempt: Patient-Level Data

Our first investigation used patient demographics and basic visit features. Despite reasonable modeling, predictive power was extremely weak, yielding $R^2 \approx 0.008$ on validation and a negative R^2 on test data.



Valuable Lessons from Initial Failure



Wait times are complex,
driven by operational factors.



Demographics alone are poor
predictors of ER congestion.



Initial findings highlighted
the need for richer context.



Critical thinking was essential
to interpret weak results.

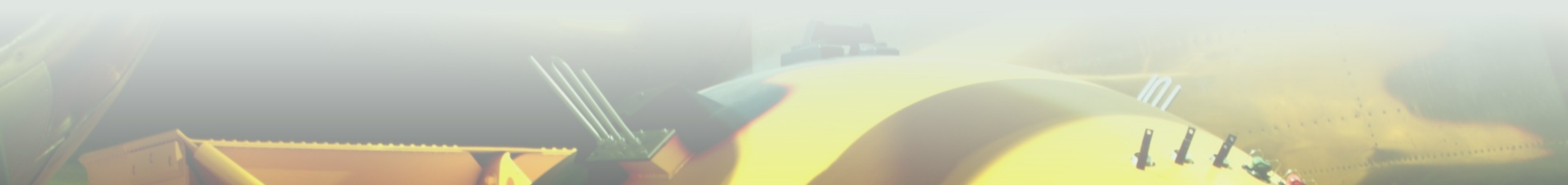


New Data, New Hope: Operational Context

Introducing a richer operational dataset dramatically changed our results. Initial models, like Linear Regression, achieved a seemingly perfect R^2 of 1.00. This near-perfect performance raised immediate red flags.



The Leakage Trap: Perfect Scores Can Lie



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We discovered target leakage: features like 'time-to-registration' indirectly encoded the target wait time. This invalidated our initial perfect results, revealing a common pitfall in predictive modeling.



Corrected Model: Trustworthy Performance

Gradient Boosting with leakage-free features

$R^2 = 0.846$

Strong predictive
power after
leakage removal

**MAE =
19.27**

Mean absolute
error in minutes

**RMSE =
26.73**

Root mean
squared error

Validated

Robust results
from operational
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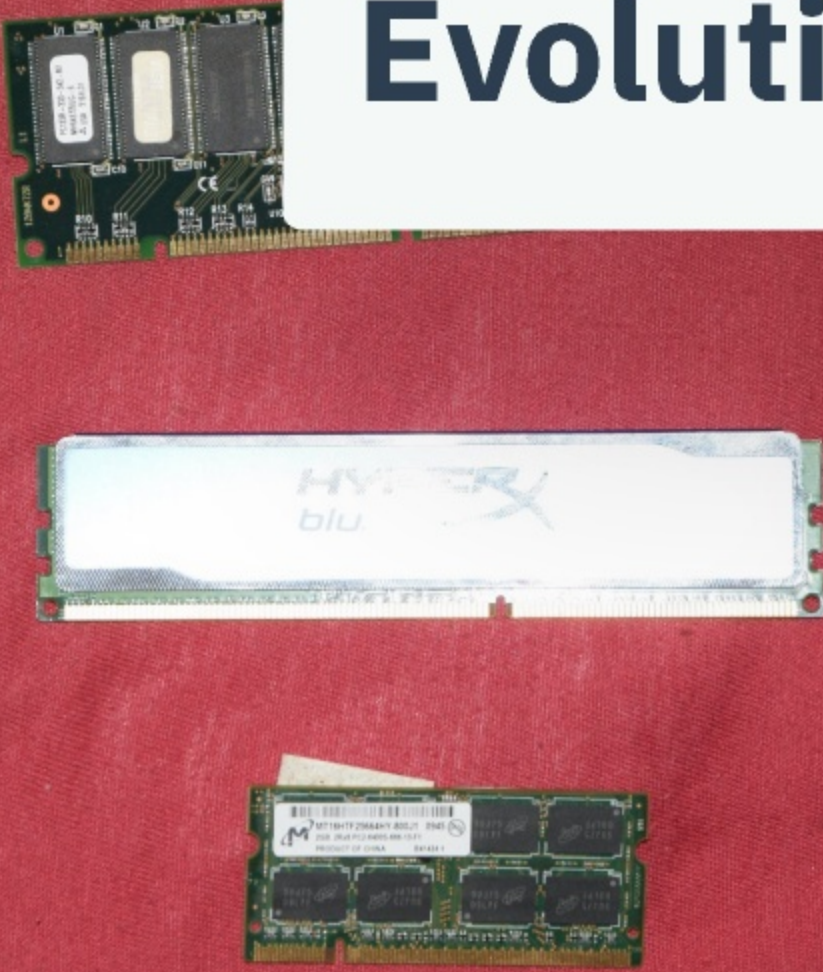
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Evolution of Predictive Power



Weak Demographic Model

$R^2 \approx 0.008$. Simple patient data offered almost no predictive value for ER wait times.

Robust Operational Model

$R^2 = 0.846$. Operational context dramatically improved accuracy, proving its relevance.

Technical Insights & Project Evaluation

- ➔ Inconsistent validation design in first notebook.
- ➔ Preprocessing must be integrated into pipelines.
- ➔ Systematic leakage checks are critical.
- ➔ CV and HPT improvements needed.
- ➔ Partial achievement of original goals (Phase 1, richer data).
- ➔ Multi-dataset stacking not fully completed.

Key Learnings & Critical Reflections



Weak results still provide valuable insights.



Feature engineering trumps blind model selection.



Leakage completely misleads evaluation.



Trustworthy ML needs rigor + domain knowledge.



Embrace iterative discovery and correction.

Conclusion: Trustworthy ER Prediction

Predicting ER wait times is possible with meaningful operational context and rigorous leakage control. Future work includes time-series CV, stronger baselines, and advanced workflow features. 'The best model is not the flashiest score—it is the one we can trust.'



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